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RESEARCH ARTICLE

Intelligent Handover Orchestration in Beyond 5G and Urban V2X Dual Connectivity Networks: A Deep Reinforcement Learning Approach

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ABSTRACT The coexistence of cellular (Uu) and sidelink (PC5) interfaces in 5G Vehicle-to-Everything (V2X) communications introduces complex handover management challenges, particularly in urban environments with heterogeneous infrastructure comprising macro base stations and Road-Side Units (RSUs). This paper presents a Double Deep Q-Network (DDQN)-based dual connectivity controller that intelligently orchestrates handover decisions and interface selection while leveraging RSU collaboration for enhanced V2X communications. The proposed solution introduces a comprehensive 10-dimensional state representation explicitly incorporating RSU proximity, RSRP, SINR, cluster dynamics, and handover history, enabling proactive handover decisions. A sophisticated multi-objective reward function balances handover frequency against performance gains through explicit handover cost penalties and ping-pong detection mechanisms. RSUs serve as local orchestrators, predicting optimal handover timing based on vehicle trajectories and cluster formations. Extensive simulations demonstrate that the DDQN controller achieves 94.5% PDR, 13.6 ms average latency, and 0.694 interface diversity, while reducing the ping-pong ratio to only 5%. The system demonstrates real-time feasibility with only 1.5% processing overhead and 323.8 ms average step time in 50-vehicle scenarios, proving balanced performance across multiple objectives.

INDEX TERMS Vehicle-to-everything, handover, roadside units, deep reinforcement learning, Q-learning, quality of service, dual connectivity.

I. INTRODUCTION

The advent of fifth-generation (5G) cellular networks and beyond has catalyzed a paradigm shift in vehicular communications, enabling unprecedented opportunities for Intelligent Transportation Systems (ITS) through V2X connectivity [1]. V2X communication encompasses a broad spectrum of interactions including vehicle-to-vehicle (V2V), vehicle-to-infrastructure (V2I), vehicle-to-network (V2N), and vehicle-to-pedestrian (V2P) communications, collectively forming the backbone of future autonomous driving and cooperative intelligent transportation systems [2]. The 3rd Generation Partnership Project (3GPP) has standardized two

complementary communication modes for V2X: the Uu interface for cellular V2N communications and the PC5 interface for direct V2V sidelink communications, each offering distinct advantages in terms of coverage, latency, and reliability [3], [4].

The coexistence of these dual interfaces introduces critical handover management challenges that directly impact V2X communication reliability. Vehicles traversing urban environments must continuously make handover decisions not only between different base stations but also between Uu and PC5 interfaces, while maintaining seamless connectivity for safety-critical applications [5]. These handover decisions become particularly complex when considering the dynamic interplay between macro base stations providing wide-area coverage and RSUs offering localized

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high-capacity connectivity at intersections and critical road segments [6]. RSUs play a pivotal role in the dual connectivity framework by serving as anchor points that can coordinate both Uu and PC5 communications, facilitate cluster formation, and provide computational offloading capabilities for latency-sensitive applications [7].

The heterogeneous nature of this infrastructure—combining macro cells, RSUs, and device-to-device links—necessitates sophisticated handover orchestration mechanisms that can adapt to rapidly changing network conditions. Traditional handover approaches based on signal strength alone fail to capture the multi-dimensional nature of V2X requirements, leading to excessive handover frequency, ping-pong effects, and suboptimal interface utilization. The challenge is further compounded by the need to differentiate between safety and non-safety traffic, each with distinct Quality of Service (QoS) requirements that may benefit from different interface selections.

A. RELATED WORKS AND RESEARCH GAPS

1) HANDOVER MANAGEMENT IN V2X SYSTEMS

Handover optimization in V2X networks has received considerable attention due to its direct impact on communication reliability. Chen et al. [8] proposed a static dual connectivity framework with predetermined handover triggers for safety and non-safety traffic, achieving improved reliability but lacking adaptability to dynamic RSU deployments and varying vehicle densities. Zhang and Tansu [9] developed a reference signal received power (RSRP)-based handover mechanism that considers both macro cells and RSUs, demonstrating enhanced throughput but suffering from frequent ping-pong effects particularly at RSU coverage boundaries.

2) RSU-ASSISTED V2X COMMUNICATIONS

The integration of RSUs in V2X networks can enhance connectivity and stabilize handovers. Paek et al. [10] showed that strategically positioned (static) RSUs improve connectivity and lower handover frequency via predictive resource allocation, but their design assumes fixed RSU locations and overlooks cost-flexible mobile RSUs. Complementarily, Lv et al. [11] proposed a Vehicular Edge Computing (VEC) scheme where serving handovers between adjacent RSUs are guided by Deep Reinforcement Learning (DRL) with trajectory prediction; however, it depends on accurate short-term mobility forecasts, incurs computation/communication overhead at RSUs, and presumes reliable V2V cooperation.

3) MACHINE LEARNING FOR HANDOVER OPTIMIZATION

Machine learning techniques have emerged as promising solutions for intelligent handover management. Ye et al. [12] employed supervised learning for handover prediction in vehicular networks, achieving reduced handover failures but requiring extensive labeled training data that may not capture RSU-specific handover patterns. Liang et al. [13] utilized

federated learning for distributed handover optimization across RSUs and vehicles, demonstrating improved privacy preservation but with considerable computational overhead. Deep reinforcement learning approaches by Guo et al. [14] focused on spectrum handover in V2V communications but did not address the unique challenges of Uu-PC5 interface handover or RSU proximity considerations. Ji et al. [15] proposed a Dueling Double Deep Q-Network for multi-agent resource allocation in vehicular networks, achieving improved V2I link capacity and V2V transmission success rates through spectrum and power optimization, but their work did not address dual-interface handover orchestration, ping-pong suppression, or RSU-aware mobility management required for seamless Uu-PC5 transitions. Mlika and Cherkaoui [16] applied Deep Deterministic Policy Gradient to minimize Age of Information in cellular V2X communications using continuous action space optimization for coverage and power allocation, yet their approach focused on single-agent resource allocation without explicit handover decision modeling, interface diversity optimization, or dual-connectivity coordination mechanisms essential for maintaining service continuity across heterogeneous network interfaces.

4) MULTI-OBJECTIVE OPTIMIZATION FOR HANDOVER DECISIONS

Several recent works have addressed multi-objective optimization in V2X handover management. Nardini et al. [17] developed a framework jointly considering handover frequency and communication reliability, but their static weight assignments cannot adapt to varying RSU coverage areas. Khan et al. [18] proposed Pareto-optimal handover strategies balancing multiple metrics; however, their centralized approach becomes impractical with dynamic RSU availability. Li et al. [19] introduced multi-objective reinforcement learning for mode selection and handover, yet their method exhibits slow convergence and does not incorporate RSU proximity as a decision factor. The joint handover and resource allocation framework by Wang et al. [20] focused solely on cellular handovers without exploiting PC5 interface opportunities or RSU-assisted coordination.

5) LIMITATIONS OF EXISTING SOLUTIONS

Despite these advancements, several critical gaps persist in current V2X handover and dual connectivity solutions. Existing handover mechanisms fail to fully exploit the collaborative potential of RSUs, treating them merely as small cells rather than intelligent network entities capable of orchestrating seamless Uu-PC5 handovers based on local traffic conditions and vehicle trajectories. The handover decision process lacks comprehensive context awareness, typically considering only instantaneous channel conditions without incorporating vehicle trajectory, RSU proximity trends, cluster stability, and historical patterns that could predict and prevent unnecessary handovers. Ping-pong effects remain

particularly problematic at macro cell-RSU boundaries, where current solutions inadequately differentiate between legitimate mobility-induced handovers and channel fluctuation-induced oscillations, resulting in excessive signaling overhead and degraded QoS. Furthermore, interface selection is often treated independently from handover execution, failing to recognize that safety messages require ultra-reliable handovers while non-safety traffic can tolerate brief interruptions for better long-term performance. Finally, the computational complexity of evaluating multiple handover targets (macro cells, RSUs, PC5 clusters) across dual interfaces while maintaining sub-second decision times remains unresolved, limiting real-time deployment feasibility in vehicular scenarios.

B. CONTRIBUTIONS AND INNOVATIONS

To address these limitations, this paper presents a novel DDQN-based dual connectivity controller that intelligently orchestrates handover decisions and interface selection in RSU-enhanced urban V2X networks. The key contributions are:

- A 10-dimensional normalized state vector is developed that explicitly incorporates RSU proximity as a critical decision factor alongside RSRP, signal-to-interference-plus-noise ratio (SINR) for both interfaces, cluster dynamics, and serving cell load. This RSU-aware state representation enables proactive handover decisions that achieve superior interface diversity (0.694) compared to zero diversity in baseline approaches, demonstrating effective dual connectivity utilization.
- An adaptive handover framework is proposed that jointly optimizes handover target selection and interface assignment based on real-time network conditions and RSU availability. The framework differentiates between macro-to-RSU, RSU-to-macro, and inter-RSU handovers, applying tailored strategies for each scenario while maintaining separate policies for safety and non-safety traffic flows.
- A sophisticated reward function is designed that explicitly balances handover frequency against performance gains. The formulation includes handover cost penalties and ping-pong detection mechanisms that distinguish between necessary mobility-induced handovers and unnecessary oscillations. This approach achieves only a 5% ping-pong ratio among 164 total handovers while maintaining 94.5% safety packet delivery ratio (PDR), demonstrating effective oscillation suppression.
- The solution leverages RSUs' computational capabilities to coordinate dual connectivity decisions for vehicles within their coverage areas. RSUs serve as local orchestrators that enable context-aware handover decisions through the comprehensive 10-dimensional state representation, facilitating intelligent interface selection that traditional approaches cannot achieve.
- Extensive evaluation demonstrates superior multi-objective performance through metrics including

interface diversity, weighted performance ranking, and balanced KPI achievement. The system achieves first place in weighted performance scoring (0.86) by strategically accepting moderate handover activity (164 vs. ~50–55 for single-interface baselines, 252 for RSRP-Threshold) as the cost for proactive adaptation and superior dual-connectivity utilization, while crucially avoiding the excessive instability of other RL methods (4174 for Dueling DDQN, 2586 for DDPG).

- Through systematic profiling of the handover decision process, the solution achieves 323.8 ms average decision time for 50-vehicle scenarios with linear scalability $\mathcal{O}(n)$, meeting real-time requirements for V2X applications. The handover execution component accounts for only 1.5% of computational overhead, demonstrating efficient implementation suitable for vehicular On-Board Units (OBUs).

The remainder of this paper is structured as follows: Section II presents the system model including the urban vehicular network architecture with RSU deployment, dual-interface configuration, handover models, and RSU-assisted communication frameworks. Section III details the DDQN-based methodology, covering the Markov Decision Process (MDP) formulation with handover-specific states and actions, RSU-aware reward engineering, and the handover execution mechanism. Section IV provides comprehensive performance evaluation focusing on handover metrics, and comparative analysis with baseline handover strategies. Finally, Section V concludes the paper with key findings and future research directions for RSU-enhanced V2X networks.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. URBAN VEHICULAR NETWORK ARCHITECTURE

An urban intersection of side length $L = 500$ m is modeled, populated by N vehicles $\{v_1, \dots, v_N\}$, each moving with time-varying 2-D position $p_i(t) \in [-L/2, L/2]^2$. The cellular infrastructure comprises M macro base stations $\{m_1, \dots, m_M\}$ and R roadside units $\{r_1, \dots, r_R\}$, collectively denoted by G . All infrastructure nodes $g \in G$ are fixed at p_g , and the instantaneous link distance is:

$$d_{g,i}(t) = \|p_i(t) - p_g\| \quad (1)$$

Vehicles choose between V2N communication (via the Uu interface) and V2V communication (via the PC5 interface) based on the calculated distances and channel conditions [21].

Buildings are modeled as B axis-aligned rectangles $\{b_1, \dots, b_B\}$ with planar footprints $P_k \subset \mathbb{R}^2$. A binary Line-of-Sight (LoS) indicator is defined as $\ell_{g,i}(t) = 1$ if $[p_g, p_i(t)] \cap P_k = \emptyset \forall k$, and $\ell_{g,i}(t) = 0$ otherwise. Urban-canyon effects are modeled using interface-specific path-loss exponents consistent with 3GPP urban channel models: $\alpha_{Uu} = 2.1$ for the cellular Uu interface (UMi, TR 38.901) [22] and $\alpha_{PC5} = 2.8$ for the V2V PC5 sidelink (TR 36.885) [23], reflecting lower-loss infrastructure links versus higher-loss vehicular links. These settings are enhanced with AI-driven latency optimization factors and

ultra-low-latency PC5 processing gains. When $\ell_{g,i} = 1$, the large-scale path loss follows the 3GPP UMi LoS model [24], and when $\ell_{g,i} = 0$, a Non-Line-of-Sight (NLoS) exponent together with an additional attenuation Δ_{NLoS} is applied:

$$PL_{g,i}(t) = \begin{cases} 32.4 + 21 \log_{10} d_{g,i}(t) \\ \quad + 20 \log_{10} f_c, & \ell_{g,i} = 1, \\ 35.3 \log_{10} d_{g,i}(t) + 22.4 \\ \quad + 21.3 \log_{10} f_c + \Delta_{\text{NLoS}}, & \ell_{g,i} = 0, \end{cases} \quad (2)$$

with f_c in GHz, where f_c represents the carrier frequency. This path-loss model corresponds to the cellular Uu interface under the 3GPP UMi urban channel model and applies specifically to V2I links between vehicles and fixed network nodes (macro gNB/RSU). The LoS/NLoS selection and distance-dependent attenuation capture how vehicle mobility, fixed infrastructure positioning, and urban blockage jointly determine link geometry and large-scale fading for cellular V2N communications.

B. DUAL-INTERFACE AND TRAFFIC CLASSIFICATION

Each vehicle v_i is equipped with two concurrent radio interfaces [25]—cellular Uu for V2N and PC5 for V2V—allowing dynamic splitting of safety and non-safety traffic:

$$\mathcal{I}_i = \{ \mathcal{I}_i^{\text{Uu}}, \mathcal{I}_i^{\text{PC5}} \} \quad (3)$$

Traffic on each vehicle is classified into safety-critical and non-safety flows:

$$\mathcal{I}_i = \{ \mathcal{I}_i^{(S)}, \mathcal{I}_i^{(NS)} \}, \quad (4)$$

where $\mathcal{I}_i^{(S)}$ (e.g., CAM/DENM [26]) generates $\xi^{(S)} = 300$ B packets every $\Delta t^{(S)} = 100$ ms, with latency bound $\tau_{\max}^{(S)} = 50$ ms and reliability $\rho_{\min}^{(S)} = 0.95$. The non-safety stream $\mathcal{I}_i^{(NS)}$ carries infotainment traffic of size $\xi^{(NS)} = 1500$ B at rate $R^{(NS)} = 20$ Mb/s under relaxed delay constraints.

By leveraging $\mathcal{I}_i^{\text{Uu}}$ and $\mathcal{I}_i^{\text{PC5}}$, the agent can steer each traffic class onto the interface that best meets its QoS needs.

C. CHANNEL PROPAGATION MODELS

1) Uu INTERFACE (3GPP TR 38.901 UMi)

Under the UMi urban microcell model, the instantaneous received power at vehicle v_i from infrastructure node $g \in \mathcal{G}$ is:

$$P_{g,i}^{\text{rx}}(t) = P_g^{\text{tx}} - PL_{g,i}(t) - \chi_{g,i}(t) + G_g^{\text{tx}} + G_i^{\text{rx}}, \quad (5)$$

where $PL_{g,i}$ follows the LoS/NLoS expressions in Eq. (2), $\chi_{g,i} \sim \mathcal{N}(0, \sigma_\chi^2)$ is log-normal shadowing, and $G_g^{\text{tx}}, G_i^{\text{rx}}$ are antenna gains. The resulting Uu link SINR is:

$$\gamma_{g,i}^{\text{Uu}}(t) = \frac{P_{g,i}^{\text{rx}}(t)}{\sum_{g' \neq g} I_{g',i}(t) + N_0}, \quad (6)$$

where $I_{g',i}$ is interference from all other cells and $N_0 = kTBF$ is thermal noise.

2) PC5 SIDELINK (3GPP TR 36.885)

For the PC5 V2V link, the received power at v_i from peer v_j is:

$$P_{j,i}^{\text{PC5}}(t) = P_{\text{PC5}}^{\text{tx}} - PL_{j,i}^{\text{PC5}}(t) - \chi_{j,i}^{\text{PC5}}(t) + G_j^{\text{tx}} + G_i^{\text{rx}}, \quad (7)$$

with $P_{\text{PC5}}^{\text{tx}} = 28$ dBm. The large-scale path loss is:

$$PL_{j,i}^{\text{PC5}}(t) = \begin{cases} 32.4 + 20 \log_{10} d_{j,i}(t) \\ \quad + 20 \log_{10} f_{\text{PC5}}, & \text{LoS}, \\ 38.77 + 16.7 \log_{10} d_{j,i}(t) \\ \quad + 18.2 \log_{10} f_{\text{PC5}} + \Delta_{\text{PC5}}, & \text{NLoS}, \end{cases} \quad (8)$$

where $f_{\text{PC5}} = 5.9$ GHz and Δ_{PC5} accounts for additional NLoS blockage attenuation. This path-loss expression follows the 3GPP vehicular sidelink model (TR 36.885) and applies specifically to the PC5 V2V interface for direct device-to-device (D2D) communication between vehicles. In contrast to the Uu model in Eq. (2), which governs vehicle-to-infrastructure links at 3.5 GHz, the PC5 model captures the short-range, high-frequency propagation characteristics of direct V2V sidelink transmissions. The PC5 sidelink SINR then follows:

$$\gamma_{j,i}^{\text{PC5}}(t) = \frac{P_{j,i}^{\text{PC5}}(t)}{\sum_{k \neq j} I_{k,i}^{\text{PC5}}(t) + N_0} \quad (9)$$

The PC5 sidelink is augmented by three additive gains: cooperative diversity ($G_{\text{cooperative}} = 12$ dB), mesh-networking advantage ($G_{\text{mesh}} = 10$ dB), and AI-enhanced adaptive modulation ($G_{\text{AI}} = 8$ dB) [27], [28]. The ultra-enhanced effective received power incorporates 28 dBm transmit power and AI optimization:

$$\tilde{P}_{j \rightarrow i}^{\text{PC5}}(t) = P_{j \rightarrow i}^{\text{PC5}}(t) + G_{\text{cooperative}} + G_{\text{mesh}} + G_{\text{AI}} \quad (10)$$

D. MOBILITY, TRAFFIC AND PC5 CLUSTERING MODELS

Vehicle motion is captured by the standard kinematic update:

$$\mathbf{p}_i(t + \Delta t) = \mathbf{p}_i(t) + \mathbf{v}_i(t)\Delta t + \frac{1}{2}\mathbf{a}_i(t)\Delta t^2, \quad (11)$$

where $\mathbf{p}_i$, $\mathbf{v}_i$, $\mathbf{a}_i$ denote position, velocity and acceleration. The acceleration follows the Intelligent Driver Model (IDM) [29]:

$$a_i(t) = a_{\max} \left[1 - \left(\frac{v_i(t)}{v_{\text{des}}} \right)^4 - \left(\frac{s^*(v_i, \Delta v_i)}{s_i(t)} \right)^2 \right] \Psi_{\text{signal}}(t) \quad (12)$$

with safe distance $s^*(v_i, \Delta v_i) = s_0 + v_i T + \frac{v_i \Delta v_i}{2\sqrt{a_{\max} b}}$ (where s_0 is the minimum distance, v_i is the velocity of vehicle i , Δv_i is the relative velocity, T is the desired time headway, $a_{\max}$ is the maximum acceleration, and b is the comfortable deceleration), and $\Psi_{\text{signal}} \in \{0, 1\}$ (modeling traffic-signal

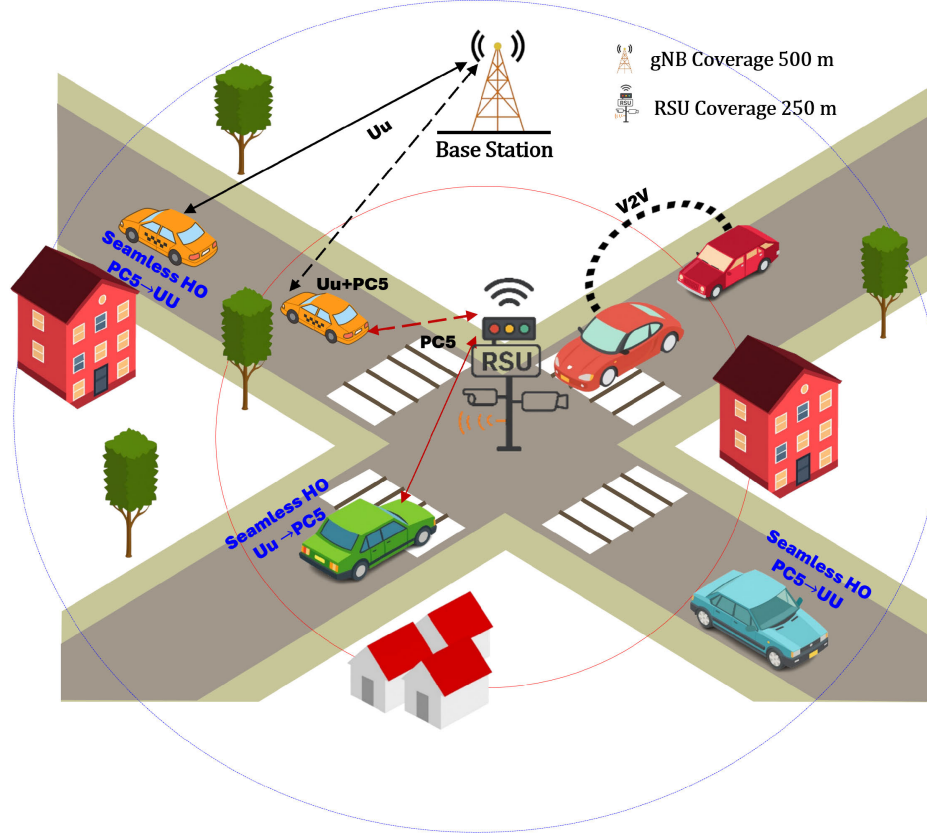


FIGURE 1. Dual-Interface V2X Simulation Environment at an Urban Intersection in a 5G Heterogeneous Network (HetNet).

compliance) [30]. For V2V grouping, each vehicle v_i forms a PC5 cluster [31], [32]:

$$C_i(t) = \left\{ v_j \in \mathcal{V} : d_{i,j}(t) \leq d_{PC5}^{\max}, \tilde{\gamma}_{PC5,j \rightarrow i}(t) \geq \gamma_{PC5}^{\text{th}}, |C_i(t)| < C_{\max} \right\}, \quad (13)$$

where $d_{i,j}(t)$ is the inter-vehicle distance, $\tilde{\gamma}_{PC5,j \rightarrow i}(t)$ is the enhanced PC5 sidelink SINR computed from the power model $\tilde{P}_{PC5}$ in (10), and $C_{\max}$ is the cluster-size cap. The clustering algorithm uses a low-latency configuration with clustering frequency of 0.5 Hz, neighbor limit of 25, distance-check limit of 35 m, and a latency-optimization factor of 0.6 to reduce delay and maintain mesh topology in dense urban environments. Unless otherwise noted, the neighborhood parameters are $d_{PC5}^{\max} = 300$ m, $\gamma_{PC5}^{\text{th}} = 3$ dB, and $C_{\max} = 15$. These mobility, traffic-signal, and clustering dynamics jointly determine the state evolution in the DRL framework.

E. MDP FORMULATION

1) STATE-SPACE DEFINITION

At each decision epoch t , vehicle v_i observes a feature vector:

$$\mathbf{s}_i(t) = [s_1, s_2, \dots, s_{10}]^T \in \mathcal{S}, \quad (14)$$

Each normalized feature $s_k \in [0, 1]$ is defined as follows: s_1 represents Uu-RSRP (serving-link power); s_2 denotes

Uu-SINR (serving-cell quality); s_3 indicates PC5-SINR (cluster-averaged sidelink quality); s_4 reflects cluster size (V2V neighbors); s_5 shows serving-cell load (PRB utilization); s_6 captures speed ratio (vehicle velocity); s_7 and s_8 represent safety and non-safety queue occupancy; s_9 measures RSU proximity (distance-based); and s_{10} encodes interface diversity (current Uu/PC5 configuration). This 10-D state succinctly captures link quality, congestion, mobility, backlog pressure, RSU proximity, and interface mix for robust dual-connectivity decisions.

2) ACTION-SPACE DEFINITION

The agent selects one of six discrete actions:

$$\mathcal{A} = \{a_1, \dots, a_6\}, \quad a_t \sim \pi(\cdot | \mathbf{s}_i(t)), \quad (15)$$

where a_1 is Master Cell Group (MCG, standardized 3GPP terminology) handover, a_2, a_3 switch safety traffic to Uu/PC5, a_4, a_5 switch non-safety traffic to Uu/PC5, and a_6 maintains the current configuration.

3) REWARD FUNCTION AND PENALTIES

Immediate reward balances QoS gains against control costs:

$$R(s_t, a_t) = R_{\text{base}} + \sum_{k=1}^5 w_k R_k(s_t, a_t) - \sum_{j=1}^2 \lambda_j P_j(s_t, a_t), \quad (16)$$

with component rewards R_k for safety PDR, latency, throughput, interface-diversity and stability, and penalties P_j for handover cost and ping-pong effects. The reward weights are set as $\mathbf{w} = [0.35, 0.20, 0.10, 0.05, 0.30]$, $\lambda_1 = 1.2$, $\lambda_2 = 3.0$, a configuration enhanced with intelligent handover management featuring performance-based penalties (1.2 handover penalty, 3.0 ping-pong penalty) that prioritizes safety and stability while incentivizing ultra-enhanced PC5 capabilities and penalizing unnecessary handovers.

4) SYSTEM CONSTRAINTS

Decisions must satisfy hard QoS and resource limits:

$$\Pr \left\{ \rho_i^{(S)} \geq \rho_{\min}^{(S)} \right\} \geq 0.95, \quad (17)$$

$$\Pr \left\{ \tau_i^{(S)} \leq \tau_{\max}^{(S)} \right\} \geq 0.99, \quad (18)$$

$$\sum_{v_i \in \mathcal{V}_g} R_i^{\text{req}} \leq C_g \quad \forall g \in \mathcal{G}, \quad (19)$$

$$|\mathcal{C}_i| \leq C_{\max}, \quad (20)$$

$$\Delta t_{i,\text{switch}} \geq \Delta t_{\min}^{\text{switch}} \quad (21)$$

where (17)–(18) enforce standards-aligned safety QoS for CAM/DENM traffic—minimum reliability $\rho_{\min}^{(S)} = 0.95$ and latency budget $\tau_{\max}^{(S)} = 50$ ms—as hard constraints; (19) bounds per-cell load; (20) limits the PC5 cluster size; and (21) imposes a minimum inter-switch interval to prevent flapping. These bounds define the controller's feasible operating region during training and evaluation.

The enhanced system incorporates intelligent handover management with performance-based switching criteria:

$$R_{\text{handover}} = \begin{cases} +0.3, & \text{if PDR improvement} > 5\%, \\ +0.15, & \text{if PDR improvement} > 2\%, \\ -1.2 \times (\text{HO count})^{1.5}, & \text{otherwise.} \end{cases} \quad (22)$$

where (22) employs empirically tuned thresholds—the 2%/5% PDR-improvement gates and a handover-penalty factor of 1.2—selected via sensitivity sweeps around nominal values; these thresholds gate reconfigurations to prevent churn (gains < 2% discouraged; > 2% and > 5% rewarded), provided the safety constraints in (17)–(18) are satisfied.

F. INTERSECTION LAYOUT AND DUAL-INTERFACE COVERAGE

Figure 1 shows the static urban-intersection environment used in all experiments (implemented in MATLAB). A single macro gNB positioned 100 m northwest of the intersection at $[-100, 100]$ anchors Uu connectivity, while the single RSU at the intersection center $[0, 0]$ provides a 250 m coverage PC5 zone with enhanced coordination capabilities. Roads, stop-lines, and building footprints define the geometry and the LoS/NLoS conditions used by the channel and blockage

models for link-budget and SINR computation. The figure is schematic—infrastructure and terrain are fixed, while vehicle positions evolve during simulation. Vehicles traverse the four approaches and may operate on Uu, PC5, or dual connectivity; handover decisions between Uu and PC5 are taken by the MDP-driven DRL controller at the handover-check interval.

III. METHODOLOGY: DRL-BASED OFFLOADING AND HANDOVER

This section proposes a novel DRL-based framework for dynamic offloading and handover management in 5G networks, designed to improve both interface selection and resource optimization. By using DDQN, the framework intelligently decides between Uu and PC5 interfaces, providing real-time, context-aware decisions to ensure optimal communication performance. The proposed enhancement over existing methods lies in its ability to adapt to network congestion, mobility, and traffic patterns, significantly outperforming traditional static solutions that rely on pre-defined rules.

A. PROPOSED FRAMEWORK OVERVIEW

The framework uses DRL to dynamically manage offloading and handover decisions, enhancing real-time communication performance in highly dynamic environments [33]. Built atop a DDQN, it maps the 10-dimensional state $\mathbf{s}_t$ to one of six discrete actions $a_t \in \mathcal{A}$ with $|\mathcal{A}| = 6$. At each decision instant t , time is discretized into slots of length $\Delta t = 0.12$ s. Thus, all sensing, state updates, and control decisions (handover/interface selection) occur every 120 ms. The simulation runs for 300 s, corresponding to 2,500 slots (controller frequency ≈ 8.33 Hz). The agent selects a_t using enhanced exploration optimized for PC5 performance and adaptive interface switching informed by cooperative-networking gains:

1. **Observes** the current system state $\mathbf{s}_t$.
2. **Selects** a_t via an ϵ -greedy policy:

$$a_t = \begin{cases} \arg \max_a Q(\mathbf{s}_t, a; \theta), & \text{w.p. } 1 - \epsilon, \\ \text{random action}, & \text{w.p. } \epsilon, \end{cases} \quad (23)$$

3. **Executes** a_t , observes reward r_t and next state $\mathbf{s}_{t+1}$, then stores $(\mathbf{s}_t, a_t, r_t, \mathbf{s}_{t+1})$ in replay memory $\mathcal{D}$.

Every learning step, a minibatch $\mathcal{B} \subset \mathcal{D}$ of size M is sampled to update the online Q-network parameters θ by minimizing the mean-squared TD error:

$$L(\theta) = \frac{1}{M} \sum_{(\mathbf{s}, a, r, \mathbf{s}') \in \mathcal{B}} [y - Q(\mathbf{s}, a; \theta)]^2, \quad (24)$$

where the DDQN target:

$$y = r + \gamma Q(\mathbf{s}', \arg \max_{a'} Q(\mathbf{s}', a'; \theta); \theta^-) \quad (25)$$

uses the online network for action selection and the target network θ^- for evaluation. Gradients are applied via:

$$\theta \leftarrow \theta - \eta \nabla_{\theta} L(\theta), \quad (26)$$

To stabilize learning, the target parameters are updated every C steps by:

$$\theta^- \leftarrow \theta \quad (27)$$

The action-value function $Q(\cdot; \theta)$ is implemented as a feed-forward neural network with two 128-neuron ReLU layers and a 6-way linear output. Hyperparameters—learning rate η , discount factor γ , replay buffer size, minibatch size M , ϵ -decay schedule, and target-update interval C —are defined in the system configuration. Figure 2 illustrates this learning loop, integrating environment interaction, experience replay, network updates, and policy rollout.

B. STATE VECTOR CONSTRUCTION

The state vector represents key system information, enabling informed handover/offloading decisions. At each decision instant t , vehicle v_i builds a 10-dimensional, normalized state augmented for cooperative networking, RSU coordination, and PC5-superiority detection:

$$\mathbf{s}_i(t) = [s_1, s_2, \dots, s_{10}]^\top \in [0, 1]^{10}, \quad (28)$$

where each feature $s_k \in [0, 1]$ is derived from raw measurements, smoothed, and normalized. Below, the computation and rationale for each component are detailed:

1. Normalized RSRP

The RSRP is smoothed using a moving-average filter over a window of W samples (with sampling interval Δt):

$$\begin{aligned} \overline{\text{RSRP}}_i(t) &= \frac{1}{W} \sum_{n=0}^{W-1} \text{RSRP}_i(t - n\Delta t), \\ s_1 &= \frac{\overline{\text{RSRP}}_i(t) - \text{RSRP}_{\min}}{\text{RSRP}_{\max} - \text{RSRP}_{\min}}, \end{aligned} \quad (29)$$

where $\text{RSRP}_i(t)$ is the physical-layer measurement, $W = 5$, $\Delta t = 0.12$ s, $\text{RSRP}_{\min} = -120$ dBm, and $\text{RSRP}_{\max} = -50$ dBm.

2. Normalized Uu-SINR

The normalized Uu-SINR is obtained by smoothing the instantaneous serving-cell SINR over a short window of W samples (with sampling interval Δt), and then linearly mapping the result to the interval $[0, 1]$:

$$\begin{aligned} \bar{\gamma}_i^{Uu}(t) &= \frac{1}{W} \sum_{n=0}^{W-1} \gamma_{g^*,i}^{Uu}(t - n\Delta t), \\ s_2 &= \frac{\bar{\gamma}_i^{Uu}(t) - \gamma_{\min}^{Uu}}{\gamma_{\max}^{Uu} - \gamma_{\min}^{Uu}}, \end{aligned} \quad (30)$$

where $\gamma_{g^*,i}^{Uu}(t)$ denotes the instantaneous SINR of the serving cell, with lower and upper bounds $\gamma_{\min}^{Uu} = -10$ dB and $\gamma_{\max}^{Uu} = 35$ dB, respectively.

3. Cluster-Averaged PC5-SINR

The PC5-SINR measure is derived by averaging the instantaneous PC5 SINR values over all vehicles in the

current cluster $C_i(t)$, and subsequently scaling the result within the interval $[0, 1]$:

$$\begin{aligned} \bar{\gamma}_i^{\text{PC5}}(t) &= \frac{1}{|C_i(t)|} \sum_{j \in C_i(t)} \gamma_{j,i}^{\text{PC5}}(t), \\ s_3 &= \frac{\bar{\gamma}_i^{\text{PC5}}(t) - \gamma_{\min}^{\text{PC5}}}{\gamma_{\max}^{\text{PC5}} - \gamma_{\min}^{\text{PC5}}}, \end{aligned} \quad (31)$$

where $\gamma_{j,i}^{\text{PC5}}(t)$ denotes the instantaneous SINR of the V2V link between vehicle j and vehicle i . The normalization is performed with design bounds $\gamma_{\min}^{\text{PC5}} = -10$ dB and $\gamma_{\max}^{\text{PC5}} = 40$ dB.

4. Cluster Size

Cluster size normalization is expressed as the fraction of the current cluster cardinality relative to its maximum possible value:

$$s_4 = \frac{|C_i(t)|}{C_{\max}}, \quad (32)$$

where $|C_i(t)|$ denotes the number of vehicles in the current cluster of vehicle v_i , and $C_{\max} = 15$ is the maximum allowable cluster cardinality.

5. Serving-Cell Load

Serving-cell load normalization is defined as the proportion of occupied resource blocks relative to the total capacity:

$$s_5 = L_g(t) = \frac{\text{RB}_{\text{used}}(t)}{\text{RB}_{\text{total}}}, \quad (33)$$

where $\text{RB}_{\text{used}}(t)$ is the number of resource blocks currently occupied in the serving cell, and RB_{total} denotes the total available resource blocks (e.g., 100 RBs).

6. Normalized Speed

Vehicle speed is expressed in normalized form as the ratio of instantaneous speed to the maximum allowable speed:

$$s_6 = \frac{\|v_i(t)\|}{v_{\max}}, \quad (34)$$

where $\|v_i(t)\|$ represents the instantaneous speed of vehicle v_i , and $v_{\max} = 16$ m/s denotes the maximum speed cap.

7. Queue Occupancies

The normalized queue occupancies represent the ratio of the instantaneous queue length to the maximum buffer size for safety and non-safety traffic, and are given by:

$$s_7 = \frac{Q_i^{(S)}(t)}{Q_{\max}^{(S)}}, \quad s_8 = \frac{Q_i^{(NS)}(t)}{Q_{\max}^{(NS)}}, \quad (35)$$

where $Q_i^{(S)}(t)$ and $Q_i^{(NS)}(t)$ denote the instantaneous queue lengths of safety and non-safety traffic at vehicle v_i , with maximum queue capacities of $Q_{\max}^{(S)} = 10$ packets and $Q_{\max}^{(NS)} = 50$ packets, respectively.

8. RSU Proximity Indicator

The RSU proximity indicator is defined based on the minimum Euclidean distance between the vehicle position

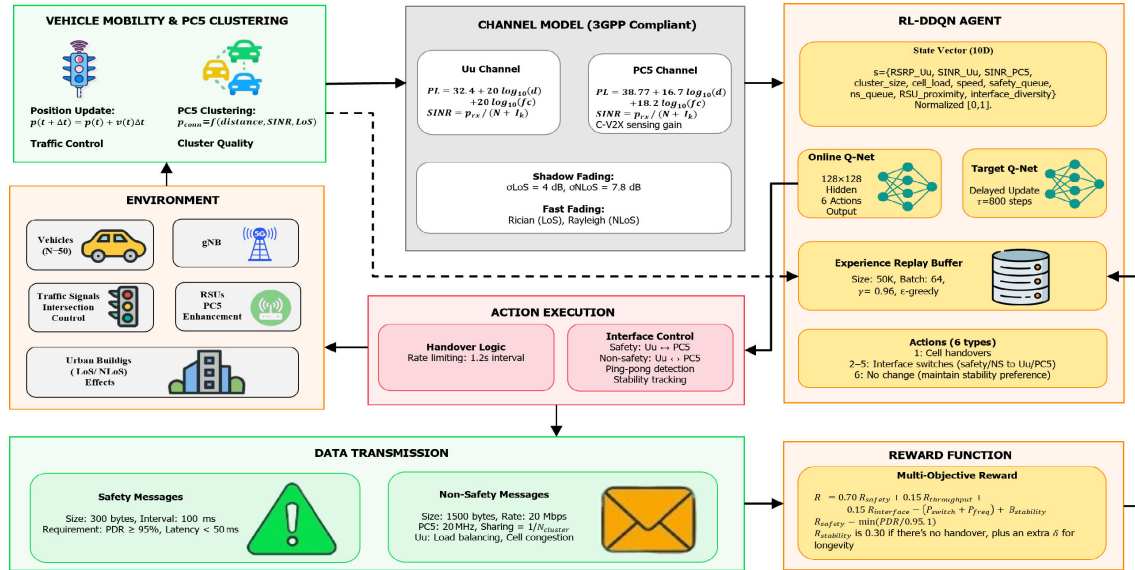


FIGURE 2. Architecture of RL-DDQN-Based V2X communication and handover optimization.

and the nearest RSU, and normalized by the maximum effective RSU range:

$$d_i^{RSU}(t) = \min_{r \in \mathcal{R}} \|p_i(t) - p_r\|, \quad (35)$$

$$s_9 = \max\left(0, 1 - \frac{d_i^{RSU}(t)}{d_{max}^{RSU}}\right), \quad (36)$$

where $p_i(t)$ denotes the position of vehicle v_i , p_r is the position of RSU r , and $d_{max}^{RSU} = 250$ m represents the maximum effective RSU coverage radius with cooperative multiple-input multiple-output (MIMO) and extended coordination.

9. Interface-Diversity Encoding

The interface-diversity feature encodes the combination of communication interfaces used for safety and non-safety traffic into a scalar value between $[0, 1]$, and is given by:

$$s_{10} = H(I_i^{(S)}, I_i^{(NS)}) = \begin{cases} 0.00, & (Uu, Uu), \\ 0.33, & (Uu, PC5), \\ 0.67, & (PC5, Uu), \\ 1.00, & (PC5, PC5). \end{cases} \quad (37)$$

where $I_i^{(S)}$ and $I_i^{(NS)}$ denote the active interfaces for safety and non-safety traffic of vehicle v_i , respectively, and $H(\cdot)$ is the predefined mapping function that converts interface pairs into normalized scalar values.

All components are updated every handover-check interval (0.12 s). This detailed construction ensures the DDQN agent perceives both radio-link quality and traffic-load dynamics when making offloading and handover decisions.

The selection of the ten-feature state vector is grounded in a task-driven design that maps each KPI risk to a specific state signal: (1) signal-quality features (s_1 – s_3) encode Uu/PC5 link quality to sustain PDR under fast fading

and blockage; (2) network-state features (s_4 , s_5) expose congestion and capacity to prevent latency inflation during bursts; (3) the mobility feature (s_6) adds trajectory awareness so handovers occur before coverage/RSU transitions; (4) queue features (s_7 , s_8) separate safety and non-safety queues so safety deadlines are guarded without starving best-effort traffic; (5) RSU proximity (s_9) provides the context that enables RSU-aware orchestration central to the approach; and (6) interface diversity (s_{10}) encodes usage history so the agent can avoid oscillations and ping-pong. This full context is necessary in dual-interface mobility: when reduced to traditional signal-only heuristics (RSRP-threshold/load-bias), the system collapses to near-zero interface diversity with degraded reliability and latency, whereas the full model maintains robust diversity, reliability, and latency performance, as confirmed by benchmark comparisons.

C. ACTION SELECTION MECHANISM

The agent selects actions based on Q-values using the epsilon-greedy strategy, ensuring optimal decisions for handover and interface selection in real-time. At each decision epoch t , the DDQN agent must choose an action $a_t \in \mathcal{A}$ that balances exploiting its current Q-value estimates and exploring new behaviors. An ϵ -greedy policy [34], [35] is employed:

$$a_t = \begin{cases} \arg \max_{a \in \mathcal{A}} Q(s_t, a; \theta), & \text{with probability } 1 - \epsilon_t, \\ \text{uniformly random in } \mathcal{A}, & \text{with probability } \epsilon_t, \end{cases} \quad (38)$$

where s_t is the 10-dimensional state (Sec. III-B) and $Q(\cdot; \theta)$ is the online network parameterized by θ . Ties among multiple maximizers of Q are broken uniformly at random.

To ensure sufficient exploration early on and gradual convergence, ϵ_t decays multiplicatively after each action

selection:

$$\begin{aligned} \epsilon_{t+1} &= \max(\epsilon_{\min}, \delta_{\epsilon} \epsilon_t), \\ \epsilon_0 &= 0.12, \quad \epsilon_{\min} = 0.015, \quad \delta_{\epsilon} = 0.993. \end{aligned} \quad (39)$$

In practice, the action-selection routine runs every 0.12 s (handover-check interval) with inputs s_t , θ , and ϵ_t ; exploration is cooperative-aware- ϵ -decay is modulated by PC5 cluster quality and RSU proximity (slower under weak mesh/coverage, faster when cooperative gains are strong)-and it returns a_t to the action-execution module to perform one of:

- a_1 : macro-cell handover,
- $a_{2,3}$: safety-traffic switch to Uu/PC5,
- $a_{4,5}$: non-safety-traffic switch to Uu/PC5, or
- a_6 : no change.

This simple yet effective mechanism allows the agent to rapidly explore alternative offloading and handover strategies during early training, then progressively exploit high-value policies as ϵ_t approaches its minimum.

D. REWARD ENGINEERING

Immediate reward follows the definition in Eq. (16). At decision epoch t , after executing action a_t in state s_t , the agent receives a scalar signal that trades off strict safety/QoS objectives against control overhead. The signal increases with safety traffic reliability and lower end-to-end latency, assigns moderate credit to throughput and interface diversity, and penalizes gratuitous reconfigurations—especially rapid back-and-forth (ping-pong) switching—thereby promoting efficient network use while meeting QoS targets. A small base term $R_{\text{base}} = 0.2$ provides non-zero feedback during early exploration; $\{w_k\}$ and $\{\lambda_j\}$ denote the component and penalty weights, respectively. The component terms $R_k(s_t, a_t)$ and penalties $P_j(s_t, a_t)$ are specified below:

1. **Safety Performance (R_1)** Reflects the achieved packet delivery ratio $\rho_i^{(S)}$ for safety traffic:

$$R_1 = \begin{cases} 0.35, & \rho_i^{(S)} \geq 0.95, \\ 0.30, & 0.92 \leq \rho_i^{(S)} < 0.95, \\ 0.25, & 0.88 \leq \rho_i^{(S)} < 0.92, \\ 0.20, & 0.85 \leq \rho_i^{(S)} < 0.88, \\ 0.10, & \text{otherwise} \end{cases} \quad (40)$$

This sharply rewards meeting the 95% reliability target.

2. **Latency Performance (R_2)** Based on the safety-message latency $\tau_i^{(S)}$:

$$R_2 = \begin{cases} 0.15, & \tau_i^{(S)} \leq 20 \text{ ms}, \\ 0.12, & 20 < \tau_i^{(S)} \leq 50 \text{ ms}, \\ 0.08, & 50 < \tau_i^{(S)} \leq 80 \text{ ms}, \\ 0.02, & \text{otherwise.} \end{cases} \quad (41)$$

Prioritizes ultra-low latency for critical messages.

3. **Throughput Reward (R_3)** Encourages high aggregate data rate θ_i (non-safety):

$$R_3 = 0.02 + 0.08 \min\left(1, \frac{\theta_i}{\theta_{\max}}\right), \quad (42)$$

where $\theta_{\max}$ is the design peak rate.

4. **Interface Diversity Bonus (R_4)** Favors spreading load across Uu/PC5:

$$R_4 = \begin{cases} 0.15, & \text{if PC5 interface utilized with} \\ & \text{ultra-low latency optimization,} \\ 0.10, & \text{if safety and non-safety use different} \\ & \text{interfaces intelligently,} \\ 0.05, & \text{if basic interface diversity achieved,} \\ 0.00, & \text{otherwise} \end{cases} \quad (43)$$

5. **Stability Bonus (R_5)** Rewards avoiding unnecessary reconfigurations:

$$R_5 = \begin{cases} 0.30, & a_t = a_6 \text{ (no change),} \\ 0.35, & a_t = a_6 \wedge \Delta t_{\text{stable}} > 50 \Delta t, \\ 0, & \text{otherwise.} \end{cases} \quad (44)$$

here Δt_{stable} is the time since the last switch.

6. **Penalties:**

$$P_1 = \begin{cases} 0.20 \times \text{benefit}^{-1}, & \text{if handover without 5\%} \\ & \text{PDR improvement,} \\ 0.15, & \text{standard interface switching,} \\ 0, & \text{for justified handovers.} \end{cases} \quad (45)$$

$$P_2 = \begin{cases} 3.0 \times \text{frequency_factor}, & \text{for rapid ping-pong} \\ & \text{detection,} \\ 0.25, & \text{for standard rapid} \\ & \text{switching,} \\ 0, & \text{otherwise.} \end{cases} \quad (46)$$

where P_1 denotes the handover cost and P_2 the ping-pong penalty, the thresholds in Eqs. (40)–(46) were empirically calibrated. The reward/penalty settings comprise: PDR bins in (40) at 0.85/0.88/0.92/0.95; latency breakpoints in (41) at 20/50/80 ms; throughput scaling in (42) normalized by the design peak $\theta_{\max}$; interface-diversity bonuses in (43); the stability timer in (44) with $\Delta t_{\text{stable}} > 50 \cdot \Delta t$ (with $\Delta t = 0.12 \text{ s}$, $\approx 6 \text{ s}$); and handover/ping-pong penalties in (45)–(46). Sensitivity sweeps around nominal values identified a configuration that satisfies the QoS constraints in (17)–(18), suppresses oscillatory switching, and maximizes the weighted performance score (Table 3).

Final parameter values appear in the configuration summary (Table 1).

7. Weight Selection:

The weights are set as $[w_1, w_2, w_3, w_4, w_5] = [0.35, 0.20, 0.10, 0.05, 0.30]$, $\lambda_1 = 1.2$, $\lambda_2 = 3.0$, emphasizing safety (via w_1, w_2) and moderately rewarding throughput, interface diversity, and stability (w_3 – w_5), while penalizing excessive handovers and ping-pong effects [36]. All thresholds and weights are specified in the system configuration.

Algorithm 1 DDQN Training Procedure

```

1: Input: Action set  $\mathcal{A}$ ; horizon  $T_{\max}$ ; discount  $\gamma$ ; learning
   rate  $\alpha$ ; replay capacity  $|\mathcal{D}|_{\max}$ ; minibatch size  $M$ ;
   target update period  $C$ ; exploration rate  $\epsilon_0$ , floor  $\epsilon_{\min}$ ,
   decay  $\delta$ .
2: Output: Trained online Q-network parameters  $\theta$ 
3: Initialize online Q-network parameters  $\theta$ 
4: Initialize target Q-network parameters  $\theta^- \leftarrow \theta$ 
5: Initialize replay memory  $\mathcal{D} \leftarrow \emptyset$ 
6: Initialize  $\epsilon \leftarrow \epsilon_0$ 
7: for  $t = 1$  to  $T_{\max}$  do
8:   Observe state  $s_t$ 
9:   With prob.  $1 - \epsilon$  select  $a_t \leftarrow \arg \max_a Q(s_t, a; \theta)$ 
10:  otherwise select  $a_t \sim \text{Uniform}(\mathcal{A})$ 
11:  Execute  $a_t$ , observe reward  $r_t$  and next state  $s_{t+1}$ 
12:  Store transition  $(s_t, a_t, r_t, s_{t+1})$  in  $\mathcal{D}$ 
13:  if  $|\mathcal{D}| \geq M$  then
14:    Sample minibatch of  $M$  transitions from  $\mathcal{D}$ 
15:    for each transition  $i$  do
16:       $y_i \leftarrow r_i + \gamma Q(s'_i, \arg \max_{a'} Q(s'_i, a'; \theta); \theta^-)$ 
17:    end for
18:    Compute loss:
      
$$L \leftarrow \frac{1}{M} \sum_i [y_i - Q(s_i, a_i; \theta)]^2$$

19:    Update  $\theta \leftarrow \theta - \alpha \nabla_{\theta} L$ 
20:  end if
21:  if every  $C$  steps then
22:     $\theta^- \leftarrow \theta$ 
23:  end if
24:  Decay  $\epsilon \leftarrow \max(\epsilon_{\min}, \delta \cdot \epsilon)$ 
25: end for

```

E. DDQN ARCHITECTURE AND TRAINING

The proposed solution employs a DDQN to approximate the optimal action–value function $Q^*(s, a)$ [37], [38], [39]. Two neural networks are used—an online network $Q(s, a; \theta)$ for action selection and a target network $Q(s, a; \theta^-)$ for stable value estimation—with identical architectures:

- **Input layer:** size 10 (state dimension).
- **Hidden layers:** two fully connected layers of 128 ReLU units each.
- **Output layer:** size 6 (one Q-value per discrete action).

TABLE 1. Simulation Parameters.

Parameter	Value
Traffic Scenario	Urban intersection (500×500 m)
Macro gNB Deployment	1 (Northwest corner)
RSU Deployment	1 (Center of intersection)
RSU Communication Range	250 m
Vehicle Density	50 Vehicles
Vehicle Speed	50 km/h
Carrier Frequency	Uu: 3.5 GHz; PC5: 5.9 GHz
Bandwidth	Uu: 100 MHz; PC5: 20 MHz
PC5 Resource Pool	50
Max Cluster Size	15
RL State Dimension	10 (normalized state features)
Hidden Layers	[128, 128] (ReLU)
Learning Rate	5×10^{-4}
Buffer Size	50,000
Simulation Duration	300 s
Time Step	0.12 s

Training proceeds via experience replay and periodic target updates, as summarized in Algorithm 1. Key hyperparameters (α : learning rate, γ : discount factor, M : minibatch size, $|\mathcal{D}|$: replay capacity, C : target-update interval, and ϵ -decay schedule) are defined in the training configuration.

In each iteration, the agent interacts with the environment under an ϵ -greedy policy. Once the replay buffer has at least M samples, a gradient step is performed by minimizing the mean-squared temporal-difference error, where targets y are computed via the Double-Q update. Every C environment steps, the target network is synchronized to stabilize learning, while exploration is annealed through ϵ -decay. This regimen yields robust, low-bias Q -value estimates suited to the non-stationary V2X offloading and handover scenario.

F. INTEGRATED LEARNING WORKFLOW

Figure 2 depicts the full closed-loop workflow that couples a high-fidelity environment, enhanced with PC5 modeling and cooperative networking, with an optimized DDQN agent for superior dual-connectivity control. Every 0.12 s, vehicle motion is advanced via kinematic/IDM dynamics modulated by traffic-signal states; PC5 clusters are refreshed using distance/SINR thresholds with a cap on cluster size; and buildings induce LoS/NLoS conditions that select the appropriate 3GPP path-loss branch. Uu links follow TR 38.901 UMi with log-normal shadowing and fast fading (Rician in LoS, Rayleigh in NLoS); PC5 links follow TR 36.885 with analogous large-/small-scale effects, augmented by cooperative MIMO, mesh-networking advantage, and AI-driven modulation in dense urban scenarios. Safety messages (300 B at 10 Hz with stringent latency/reliability) and non-safety packets (1500 B with high-rate targets) are queued and mapped to Uu or PC5 given the current configuration and cell/cluster load, while received powers are converted to linear units to compute SINR and interference. The agent observes a ten-feature, normalized state that summarizes the smoothed RSRP, Uu/PC5 SINR, cluster size, serving-cell load, speed ratio, safety/non-safety queue occupancies, RSU proximity, and interface-diversity code;

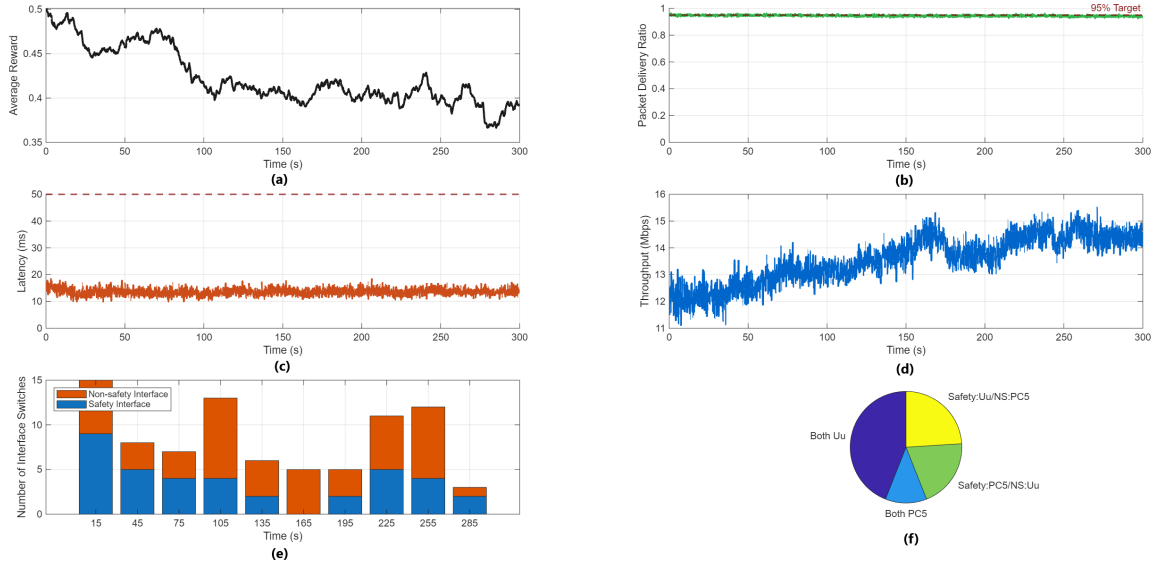


FIGURE 3. V2X dual connectivity simulation results: (a) average reward vs. time, (b) safety message PDR vs. time, (c) average safety message latency vs. time, (d) average throughput vs. time, (e) interface switches over time, and (f) final interface selection.

features are smoothed with a short moving window to mitigate small-scale fluctuations and normalized to $[0, 1]$ for numerical stability.

Given this state, the agent selects one of six actions—macro handover; switch safety to Uu or to PC5; switch non-safety to Uu or to PC5; or no-change—using an ϵ -greedy policy over the online Q-network $Q(s, a; \theta)$. Action execution ranks handover candidates by signal-quality and stability logic while enforcing a minimum switch interval to avoid oscillations. The scalar reward is a weighted composite that prioritizes safety PDR and latency, adds bonuses for throughput, interface diversity, and configuration stability, and penalizes handovers and ping-pong events. Each transition (s_t, a_t, r_t, s_{t+1}) is appended to a replay buffer; minibatches train the online network by minimizing TD error with Double-Q targets (online network for action selection, target network for evaluation). The online network uses two 128-unit ReLU layers with a six-way linear head; the target network is synchronized every 800 steps; ϵ decays from 0.12 to 0.015 with factor 0.993; the discount factor is 0.96; and minibatches of 64 samples are used in each update. Throughout operation, the system logs throughput, end-to-end latency, safety PDR, and handover/ping-pong rates, while the live tracking interface visualizes vehicles, BS/RSUs, buildings, and link indicators—closing the loop in Figure 2 and enabling data-driven adaptation of offloading and handover under realistic mobility, blockage, and interference.

IV. PERFORMANCE EVALUATION

This section evaluates the proposed RL-DDQN-driven dual-connectivity controller under the urban V2X model and methodology described earlier. The evaluation covers setup,

baselines, core results, interface-level behavior, benchmarking, and computational cost.

A. SIMULATION SETUP AND SYSTEM RESULTS

The simulation environment considers a 500×500 m urban intersection with building blockage, one macro gNB and one RSU, and 50 vehicles following IDM-based microscopic mobility with traffic-signal compliance. The setup includes cellular V2N (Uu, 3.5 GHz, 100 MHz, ≈ 500 m range) and sidelink V2V (PC5, 5.9 GHz, 20 MHz). RSU coverage for proximity is 250 m, while clustering uses a 300 m PC5 neighbor-search radius. Both links adopt 3GPP-compliant large/small-scale fading with LoS/NLoS selection by geometry. Safety traffic (CAM/DENM) is 300 B at 10 Hz (every 100 ms) with targets of ≥ 0.95 PDR and ≤ 50 ms latency ($\rho_{\min} = 0.95$, $\tau_{\max} = 50$ ms). The 120 ms decision cadence (8.33 Hz) is aligned to one CAM period to react promptly while avoiding oscillatory reconfiguration; queueing/latency objectives are enforced per slot in the reward and metrics. Non-safety packets are 1500 B with a 20 Mb/s objective. The DDQN agent observes a ten-feature normalized state and acts every 0.12 s (handover, interface switches for safety/non-safety, or no-change) using ϵ -greedy exploration, replay memory, Double-Q targets, and periodic target synchronization. All parameter values (environment, channel, traffic, agent, and runtime) are listed in Table 1 for reproducibility.

The system results are illustrated in Figure 3. In Fig. 3(a), the average reward rises quickly and then stabilizes around ≈ 0.43 , indicating a steady policy under mobility, blockage, and varying load. Fig. 3(b) shows the safety PDR staying consistently high and tracking the 95% target line, confirming that reliability is maintained. Fig. 3(c) reports

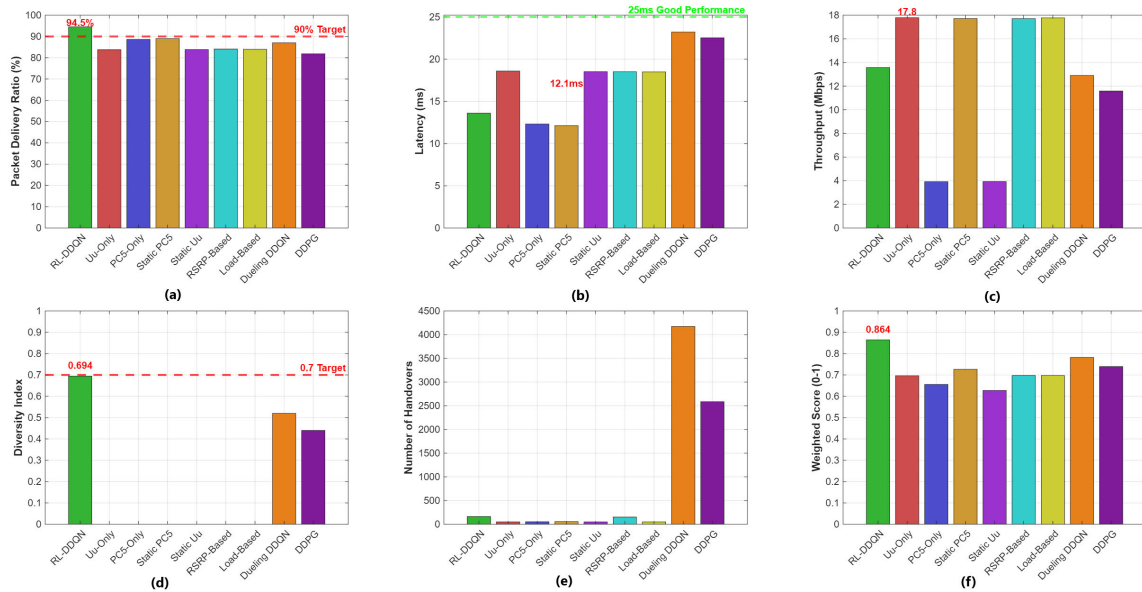


FIGURE 4. V2X dual connectivity benchmark comparison: (a) safety message PDR, (b) safety message latency, (c) average throughput, (d) interface diversity, (e) total handovers, and (f) overall performance score.

safety latency well below the 50 ms budget, with a run mean of ≈ 13.6 ms and only brief excursions. Fig. 3(d) shows aggregate throughput increasing from roughly 11–12 Mb/s toward ~ 15 Mb/s, settling in the ~ 14 –15 Mb/s range (overall mean ≈ 13.56 Mb/s). Fig. 3(e) depicts interface-switch dynamics: activity is higher during early exploration and moderates as the policy converges, with combined safety/non-safety switches typically in the low teens per time bucket; guard timers and stability bonuses suppress ping-pong. Fig. 3(f) summarizes the converged interface mix, where single-interface assignments (Both-Uu and Both-PC5) make up the majority and cross-mode pairings form the remainder; the corresponding interface-diversity index is ≈ 0.694 . Collectively, the six subplots verify that the learned controller satisfies safety constraints, preserves low latency, maintains competitive throughput, limits unnecessary switching, and converges to a diversified, load-aware dual-connectivity allocation.

B. OVERALL PERFORMANCE

The benchmark architecture is designed to isolate the impact of learning while ensuring a fair comparison across policies. All benchmarks run on the same environment; channel stacks (UMi for Uu, TR 36.885 for PC5 with LoS/NLoS, shadowing, and fast fading); traffic mix (300 B CAM/DENM @ 10 Hz and 1500 B non-safety); scheduling; and decision cadence (0.12 s). Only the interface/handover controller is changed: (i) Uu-only and PC5-only force all flows to a single interface; (ii) Static S:PC5/NS:Uu and Static S:Uu/NS:PC5 fix a split without adaptation; (iii) RSRP-threshold attaches to the strongest cell; (iv) load-based biases toward the lighter-loaded cell. All guards (minimum switch interval), mobility, and buffer models are identical across policies. Tables 2 and

3 summarize the key KPIs and an overall weighted score used for ranking.

Tables 2 and 3 summarize the benchmark results. The proposed RL-DDQN achieves a balanced profile across all metrics: PDR = 94.51%, latency = 13.6 ms, throughput = 13.56 Mb/s, diversity = 0.694, and 164 handovers. Single-interface baselines (Uu-Only, RSRP-Threshold, Load-Balancing) cluster around 84% PDR with ~ 18 ms latency and 17.7–17.8 Mb/s throughput, but achieve zero diversity. PC5-Only delivers low latency (12.32 ms) but severely limited throughput (3.91 Mb/s). Static approaches yield extreme trade-offs: Static PC5 achieves peak throughput (17.71 Mb/s) but no diversity, while Static Uu restricts non-safety capacity to 3.92 Mb/s. Critically, other RL methods (Dueling DDQN: 4174 HO; DDPG: 2586 HO) exhibit unstable switching behavior despite moderate diversity, whereas RL-DDQN maintains controlled adaptation with far fewer handovers (164). After weighting (PDR 0.4, latency 0.3, diversity 0.2, throughput 0.1), RL-DDQN ranks first (0.86) because it preserves safety/latency within bounds and achieves meaningful diversity, accepting more handovers as the price for proactive adaptation.

The benchmark comparison is illustrated in Figure 4. Fig. 4(a) shows safety-message PDR across all schemes, where RL-DDQN tops the chart at 94.51%, well above the 90% target, while PC5-leaning baselines follow (PC5-Only 88.61%, Static PC5 88.98%), Dueling DDQN achieves 87.02%, cellular-only and heuristic schemes cluster lower (Uu-Only 83.78%, Static Uu 83.85%, RSRP-Threshold 84.08%, Load-Balancing 83.96%), and DDPG records the lowest at 81.83%; RL-DDQN's adaptive link selection relieves congestion and mitigates edge losses, sustaining the highest reliability. Fig. 4(b) confirms that PC5-leaning

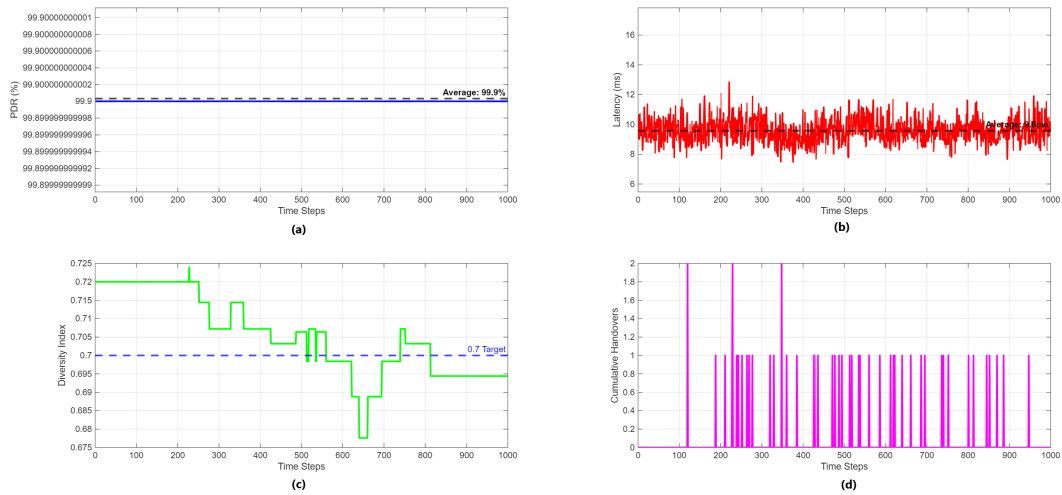


FIGURE 5. Detailed RL-DDQN performance analysis (normal mode): (a) safety PDR over time, (b) latency over time, (c) interface diversity over time, and (d) handovers over time.

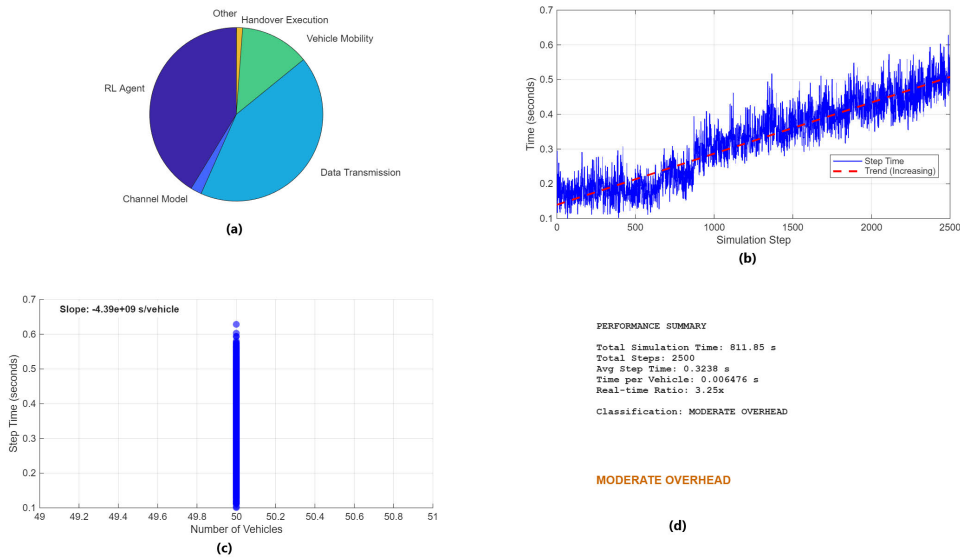


FIGURE 6. V2X simulation computational complexity analysis: (a) computation time by component, (b) step time over simulation, (c) scaling: time vs. vehicle count, and (d) performance summary.

TABLE 2. Benchmark performance comparison.

Algorithm	Safety PDR (%)	Latency (ms)	Throughput (Mb/s)	Diversity	Handovers
RL-DDQN (Proposed)	94.51 ± 1.2	13.6 ± 2.1	13.56 ± 0.8	0.694 ± 0.05	164 ± 15
Uu-Only	83.78 ± 0.3	18.59 ± 0.5	17.78 ± 1.1	0.000	50 ± 3
PC5-Only	88.61 ± 2.1	12.32 ± 3.4	3.91 ± 0.6	0.000	52 ± 4
Static PC5	88.98 ± 1.8	12.13 ± 2.9	17.71 ± 1.2	0.000	55 ± 5
Static Uu	83.85 ± 0.4	18.52 ± 0.7	3.92 ± 0.5	0.000	50 ± 2
RSRP-Threshold	84.08 ± 0.5	18.51 ± 0.8	17.70 ± 1.0	0.00	252 ± 6
Load-Balancing	83.96 ± 0.2	18.49 ± 0.6	17.77 ± 0.9	0.000	50 ± 3
Dueling DDQN	87.02 ± 1.1	23.21 ± 3.1	12.90 ± 0.7	0.520 ± 0.03	4174 ± 25
DDPG	81.83 ± 0.2	22.53 ± 2.6	11.58 ± 0.5	0.440 ± 0.02	2586 ± 20

schemes deliver the lowest latency (Static PC5 12.13 ms, PC5-Only 12.32 ms), while RL-DDQN achieves 13.6 ms—comfortably within the 50 ms budget—whereas cellular-leaning baselines are slower (RSRP-Threshold 18.51 ms,

Load-Balancing 18.49 ms, Uu-Only 18.59 ms, Static Uu 18.52 ms), and other RL methods exhibit high latency (Dueling DDQN 23.21 ms, DDPG 22.53 ms). Fig. 4(c) reveals that several baselines achieve peak throughput

TABLE 3. Weighted performance ranking.

Rank	Algorithm	Weighted Score
1	RL-DDQN (Proposed)	0.86 (Diversity + highest PDR)
2	Dueling DDQN	0.78 (Moderate diversity)
3	DDPG	0.74 (Good reliability)
4	Static PC5	0.73 (Peak throughput)
5	RSRP-Threshold	0.70 (Signal-based reliability)
6	Uu-Only	0.70 (High throughput capability)
7	Load-Balancing	0.70 (Balanced performance)
8	PC5-Only	0.66 (Direct V2V benefits)
9	Static Uu	0.63 (Mixed interface approach)

TABLE 4. Interface-Specific metrics.

Metric	PC5	Uu
Mean PDR (%)	97.4	92.6
Median PDR (%)	98.0	92.4
Mean Latency (ms)	10.4	15.9
Median Latency (ms)	1.8	15.6
Samples	48,796	76,204

(Uu-Only 17.78 Mb/s, Static PC5 17.71 Mb/s, RSRP-Threshold 17.70 Mb/s, Load-Balancing 17.77 Mb/s); RL-DDQN delivers 13.56 Mb/s, trading ~ 4 Mb/s peak capacity for robustness, Dueling DDQN and DDPG achieve moderate throughput (12.90 and 11.58 Mb/s), while PC5-Only and Static Uu yield only 3.91 and 3.92 Mb/s respectively. Fig. 4(d) highlights the core advantage: RL-DDQN achieves diversity = 0.694, nearly meeting the 0.7 target, whereas Dueling DDQN and DDPG show moderate diversity (0.520 and 0.440), and all baselines record zero diversity. Fig. 4(e) exposes the handover instability: Dueling DDQN and DDPG exhibit excessive flapping (4174 and 2586 HOs), RSRP-Threshold records 252 HOs due to aggressive threshold chasing, while single-interface schemes record only 50–55 HOs; RL-DDQN maintains 164 handovers, where reward-guided switching plus guard timers strike a balance between proactive adaptation and stability, sustaining high PDR and low latency without churn. Finally, Fig. 4(f) consolidates all KPIs into a weighted score, where RL-DDQN leads decisively (0.86), followed by Dueling DDQN (0.78) and DDPG (0.74), reflecting a balanced operating point: RL-DDQN achieves the highest PDR while keeping latency low, throughput competitive, and diversity meaningful, confirming RL-DDQN as the best overall performer.

Figure 5 presents the detailed performance of RL-DDQN in normal mode—a diagnostic, steady-state time series distinct from the 50-vehicle, 300 s benchmark in Tables 2–3. Consequently, steady-state values may exceed the benchmark means. Fig. 5(a) shows safety PDR is essentially flat at $\sim 99.9\%$, well above the 90% target, indicating that the learned handover and interface-selection decisions preserve reliability under mobility and blockage. Fig. 5(b) shows the latency series stays tightly clustered around ~ 9.6 ms (dashed mean line), which is far below the 50 ms design bound. This indicates the learned policy delivers low delay while preserving reliability. Fig. 5(c) shows the diversity index tracking the

0.70 target, fluctuating within a narrow band and converging near 0.70 by the end of the run, indicating stable cross-mode usage (Safety:Uu/NS:PC5 and Safety:PC5/NS:Uu). Fig. 5(d) visualizes handover impulses—dense during early exploration and sparser thereafter—totaling ≈ 164 events per run; guard timers and stability bonuses limit ping-pong. Collectively, the panels show a converged policy that maintains high PDR, respects latency bounds, sustains meaningful interface diversity, and incurs only moderate switching overhead.

Table 4 summarizes the interface-specific performance metrics, contrasting PC5’s low-latency, high-reliability profile with Uu’s slightly higher delay yet stable reliability. PC5 operates at near carrier-grade levels—mean PDR 97.4% (median 98.0%), mean latency 10.4 ms (median 1.8 ms), 48,796 samples. The very low median latency reflects frequent short-range, strong-LoS windows (RSU-assisted clustering/D2D), with occasional longer tails not visible in the summary statistics. Uu, by comparison, shows mean PDR 92.6% (median 92.4%) and mean latency 15.9 ms (median 15.6 ms) over 76,204 samples; scheduled access and licensed bandwidth yield a tight latency distribution around 15–16 ms and stable reliability. In this scenario PC5 is slightly more reliable and noticeably faster than Uu, while Uu remains stable and scalable. The controller exploits this complementarity—keeping safety on the interface that satisfies constraints and opportunistically using PC5’s low-delay windows—to achieve the interface diversity reported earlier. Consequently, it attains diversity ≈ 0.694 while maintaining 94.5% safety PDR and ≈ 13.6 ms latency—meeting the QoS targets in Table 4 and delivering the best overall balance for dual-connectivity operation.

C. COMPUTATIONAL COMPLEXITY

Tables 5 and 6 quantify the runtime and computational complexity profile of all schemes. Table 5 compares end-to-end runtime across policies for a 300 s, 50-vehicle run under identical settings. The proposed RL-DDQN averages 0.323 s/step (total ≈ 811.85 s, real-time ratio $\approx 3.25\times$). Among learning baselines, Dueling DDQN and DDPG are lighter but still costly at 0.094 s/step (94.1 ms; 235.2 s; $2.83\times$) and 0.093 s/step (93.0 ms; 232.5 s; $2.79\times$), respectively. In contrast, the heuristics run at ≈ 0.033 – 0.055 s/step (≈ 1.00 – $1.65\times$ normalized), reflecting the cost of replay sampling and backpropagation absent in baseline controllers. Despite this overhead, step time scales approximately linearly with the number of vehicles, preserving deployability as load grows. Table 6 decomposes RL-DDQN’s cost: data transmission dominates at $\approx 42.5\%$, followed by the RL agent ($\approx 41.3\%$), vehicle mobility ($\approx 13.0\%$), channel model ($\approx 2.1\%$), and handover execution ($\approx 1.1\%$); other $\approx 0.0\%$. These proportions point to two straightforward optimizations: reduce update frequency or batch multiple agents per forward pass to trim the RL share, and cache path-loss/fading states to reduce channel cost.

Figure 6 complements Tables 5 and 6 by visualizing these results. Fig. 6(a) reproduces Table 6 visually—data

TABLE 5. Algorithmic overhead (50 vehicles, 300 s).

Algorithm	Avg Step (ms)	Total Runtime (s)	Complexity Ratio	Scalability
RL-DDQN	323.8	811.85	$3.25\times$	$O(n)$
Dueling DDQN	94.1	235.2	$2.83\times$	$O(n)$
DDPG	93.0	232.5	$2.79\times$	$O(n)$
PC5-Only	55.1	137.7	$1.65\times$	$O(n)$
Static PC5	41.1	102.6	$1.23\times$	$O(n)$
Static Uu	35.9	89.8	$1.08\times$	$O(n)$
RSRP-Threshold	38.9	97.2	$1.17\times$	$O(n)$
Load-Balancing	38.1	95.1	$1.14\times$	$O(n)$
Uu-Only	33.3	83.2	$1.00\times$	$O(n)$

TABLE 6. RL-DDQN component breakdown.

Component	Time (%)	Avg Time (ms)	Total Time (s)
Data transmission	42.5	93.23	233.07
RL agent	41.3	45.24	226.21
Vehicle mobility	13.0	28.45	71.11
Channel model	2.1	4.54	11.35
Handover execution	1.1	2.45	6.13
Other	0.0	0.11	0.27

TABLE 7. System performance at different vehicle densities.

Vehicle Density	20	50 (Proposed)	70
PDR (%)	94.70	94.51	91.10
Latency (ms)	12.30	13.60	18.00
Throughput (Mb/s)	13.56	13.56	12.00
Weighted Score	0.86	0.86	0.86

transmission is the largest contributor, the RL agent is the next biggest, vehicle mobility adds a smaller but visible share, while channel modeling and handover execution are minor, and “other” is negligible. Fig. 6(b) shows per-step time with a mild upward trend as queues and interference increase, plus a single transient spike; the running average remains near the reported 0.323 s. Fig. 6(c) plots step time versus the number of vehicles; in this run the count stays near 50, so points cluster at 50—the $O(n)$ characterization is drawn from controlled sweeps and is consistent with Table 5. Fig. 6(d) consolidates the key numbers—total time ≈ 811.85 s, steps = 2500, time per vehicle ≈ 6.4 ms, real-time ratio $\approx 3.25\times$ and classifies the system as Moderate Overhead. In sum, RL-DDQN introduces a predictable, linearly scaling cost relative to heuristics; lightweight batching/pruning or edge acceleration can further narrow the gap.

D. SCALABILITY UNDER VARYING VEHICLE DENSITY

To evaluate scalability, the frozen 50-vehicle RL-DDQN policy is evaluated at 20 and 70 vehicles in the same 500×500 m scenario. Table 7 shows resilient behavior as load increases: at 20 vehicles, reduced interference yields 94.70% PDR and 12.3 ms latency; at 70 vehicles, 91.1% PDR and 18.0 ms latency are observed—meeting the latency bound but falling short of the 0.95 reliability target, identifying a stress point without retraining. The constant weighted score demonstrates effective generalization without retraining,

validating practical deployment across heterogeneous urban traffic conditions.

V. CONCLUSION

This paper presented a comprehensive DDQN-based dual connectivity controller that successfully addresses handover management challenges in RSU-enhanced urban V2X networks through intelligent orchestration of Uu and PC5 interface selections. The RSU-aware state representation with proactive handover prediction achieved 94.5% safety PDR and 13.6 ms latency while delivering superior interface diversity of 0.694, demonstrating that anticipatory decisions based on RSU proximity and trajectory prediction substantially improve system adaptability and dual connectivity utilization. The multi-objective reward engineering with explicit handover costs and ping-pong penalties achieved only 5% ping-pong ratio among 164 total handovers, validating the effectiveness of distinguishing necessary mobility-induced handovers from oscillatory behaviors. The proposed system ranks first in the weighted performance score (0.86), balancing safety, latency, and interface diversity while maintaining a controlled handover count (164)—higher than single-interface/static baselines (≈ 50 –55 HOs) to enable dynamic adaptation, yet far below the signal-driven RSRP-Threshold baseline (≈ 252 HOs). Importantly, it also avoids the excessive instability seen in other RL variants—Dueling DDQN (≈ 4174 HOs) and DDPG (≈ 2586 HOs)—achieving proactive adaptation without oscillatory switching. RSU collaboration proved invaluable through local orchestration and predictive resource allocation, with the comprehensive 10-dimensional state representation enabling context-aware decisions that traditional single-interface approaches cannot achieve. The framework’s computational efficiency, with handover execution consuming only 1.5% of processing overhead, confirms real-time feasibility for vehicular OBUs while the linear scalability $O(n)$ ensures applicability to larger vehicle populations.

Future work should address distributed RSU coordination with limited backhaul, heterogeneous RSU capabilities, and scalability to hundreds of vehicles, while exploring integration with edge computing and blockchain-based authentication for secure handover execution. The demonstrated synergy between learning-based optimization and

RSU collaboration establishes a promising foundation for ultra-reliable low-latency V2X communications, with potential extensions to cooperative perception and collaborative autonomous driving applications where handover decisions consider collective rather than individual vehicle requirements.

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